



Artificial Intelligence: An introduction

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Biomedical Imaging, Vision and Learning Laboratory. BivL²ab
Machine learning Analysis and Computer Vision. M_LACV
Universidad Industrial de Santander

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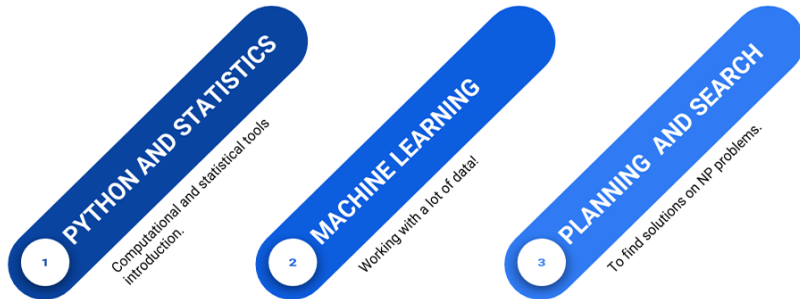


Biomedical Imaging, Vision and Learning Laboratory

Información del profesor

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- ▶ Contenido del curso:
<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

Course outline



- ▶ **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- ▶ **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course



Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

First module: Python and statistics introduction

- ▶ Introduction to Python
- ▶ Introduction to statistics
- ▶ Introduction to Visualization and data analytics



Second Module: Machine Learning

Classification and regression

- ▶ Naive Bayes
- ▶ Classification trees
- ▶ Support Vector Machines
- ▶ Introduction to Deep-learning

Non-supervised learning

- ▶ K-means
- ▶ DBScan



Third module: Planning and Searching

- ▶ Planning and searching
- ▶ Genetic algorithms (GA)
- ▶ Simulated annealing (SA)



Structure of the Course

- ▶ 30% Talleres (Problemsets)
- ▶ 10% Ejercicios clases.
- ▶ 30% Parciales (Quizes)
- ▶ 30% Proyecto funcional IA.
 - ▶ 5 % primera entrega. Exploración de datos
 - ▶ 5 % segunda entrega. Validación de modelos clase
 - ▶ 20 % Entrega final. Incluye repositorio, presentación y avance del proyecto
 - ▶ Solo se evalua el curso, si ha sido previamente aprobado antes del primer corte.

Curso en GITLAB:

<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

PREVIOUS PROJECTS

- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2018-2.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-1.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-2.md
https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2020-1.md

Some I.A. Applications: In classrooms

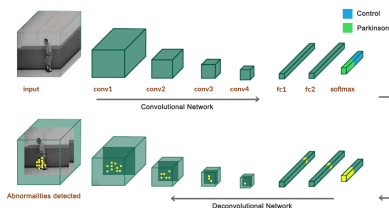


Video AI-classroom-CHINA

Some I.A. Applications: In Agriculture



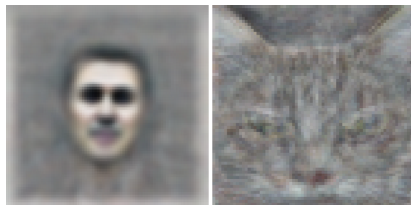
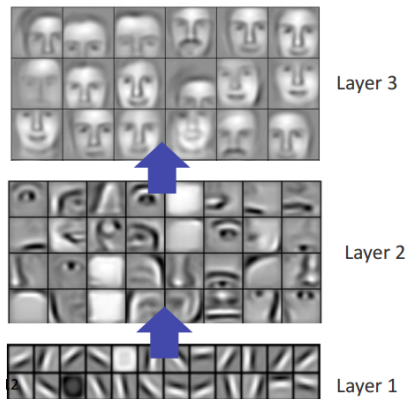
Salient Gait maps



Learning convolutional neuronal patterns

A 3D convolutional gait representation that produces salient spatio-temporal regions can potentially support and complement the diagnosis and the following of Parkinson's disease

Big challenges: How understand complex concepts?



Now, the machines can dream



A photograph of a Go match between a human player and a computer. The human player, wearing a dark suit, is seated on the left, looking at a board. The computer player, wearing a dark suit, is seated on the right, also looking at the board. A Go board is placed on a table between them. The background is dark. The text "Video 01" is overlaid in green in the upper right. The text "ALPHAGO" is overlaid in large white letters at the bottom.

Video 01

ALPHAGO

A photograph of a Go match between a human player and a computer. The human player, wearing a dark suit, is seated on the left, looking at a board. The computer player, wearing a dark suit, is seated on the right, also looking at the board. The board is a Go board with black and white stones. The background is dark. The text "Video 02" is overlaid in green. The text "ALPHAGO" is overlaid in large white letters at the bottom.

Video 02

ALPHAGO

ImageNet Challenge

IMAGENET

IMAGENET Netflix 22:30, Bill Nye

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.

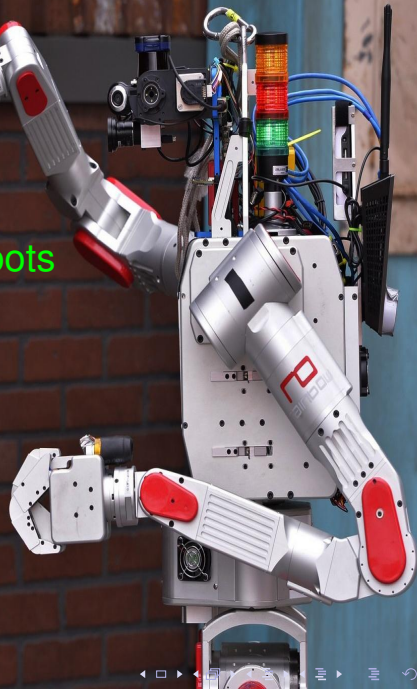


As support for decisions making

- ▶ Medicine
- ▶ Politics
- ▶ Laws



NOVA Rise of the Robots





y ahora ...Atlas. Google again!

Boston Dynamics

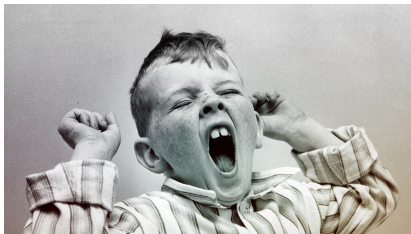
DARPA



Understanding Artificial Intelligence and Its Future

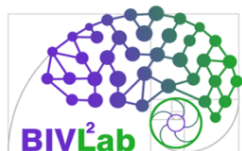


Thank you for your attention ...



... It's time to wake up

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